## Sine rule

**1**

**a** No **b** Yes **c** Yes **d** No

**2**

**a** $\frac{a}{\sin A} = \frac{b}{\sin B}$ **b** $\frac{p}{\sin P} = \frac{q}{\sin Q} = \frac{r}{\sin R}$

**3**

**a** $\sin A = \frac{x}{b}$ **b** $x = a \sin B$ **c** $\frac{\sin A}{a} = \frac{\sin B}{b}$

## Unknown sides

**4**

**a** $a = 10.50$ **b** $a = 6.85$ **c** $a = 3.02$ m **d** $a = 711.73$ cm

**e** $a = 11.29$ cm

**5**

**a** $x = 13.1$ **b** $x = 12.9$

**6**

**a**

**i** $a = 16.66$ **ii** $a = 16.66$

**b**

**i** $a = 17.14$ **ii** $a = 17.14$

**c**

**i** $a = 30.09$ **ii** $a = 30.09$

**7**

**a** $HK = 733.17$ km **b** $KJ = 444.55$ km

**8**

**a** $\angle A = 75.78$ **b** $b = 3.152$ m **c** $c = 5.723$ m

## Unknown angles

**9**

**a** $x = 26.0^\circ$ **b** $x = 32.7^\circ$ **c** $x = 50.7$ **d** $x = 30.4^\circ$

**e** $x = 20.4^\circ$ **f** $x = 30.3^\circ$ **g** $x = 43.2^\circ$ **h** $x = 149.1^\circ$

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**i** $x = 133.13^\circ$ **j** $x = 23.5^\circ$

**10**

**a** $B = 74^\circ$ **b** $B = 31^\circ$

**11**

**a** $x = 19.54^\circ$ **b** $48.46^\circ$

**12**

**a** $x = 41$ **b** $139$

**13** $\theta = 33.84^\circ$

## Applications

**14**

**a** $\angle OBA = 39^\circ$ **b** $k = 21.76$

**15** $x = 53.6 \text{ m}$

**16**

**a** $\angle A = 38^\circ$ **b** $c = 22.71 \text{ m}$ **c** $b = 22.71 \text{ m}$ **d** $1.9 \text{ min}$

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